# Micah Nye

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## Education

**Carnegie Mellon University, Robotics Institute** — Pittsburgh, PA
Ph.D. in Robotics — Expected May 2030
M.S. in Robotics *(GPA: 4.0/4.0)* — August 2026
*Advisors: Sebastian Scherer & Wenshan Wang*

**University of Pittsburgh, Swanson School of Engineering** — Pittsburgh, PA
B.S. in Mechanical Engineering *(GPA: 3.96/4.00, Dean's Honors List)* — August 2024

## Research Experience

**The AirLab, Carnegie Mellon University (🔗)** — **09/2022–Present**
*Graduate Researcher (08/2024–Present); Undergrad. Research Assistant (09/2022–08/2024)* — *Pittsburgh, PA*

- Developed perception-conditioned forward dynamics models for off-road driving, validated extensively offline and on full-scale hardware
- Co-led a 5-member team to collect a large-scale, multimodal dataset of diverse, unstructured off-road driving
- Developed and deployed sampling-based MPC and autonomy stack on wheeled, tracked, and legged embodiments using PyTorch GPU parallelization, ROS, and DDS infrastructure
- Estimated epistemic uncertainty of learned traversability costmaps for a full-scale autonomous ATV operating in unknown off-road terrain

**Indy Autonomous Challenge, MIT-PITT-RW (🔗)** — **03/2022–05/2025**
*Chief Engineer; Localization & Deployment Subteam Lead* — *Pittsburgh, PA*

- Led a 40-member, student-run team across multiple R1 institutions as Chief Engineer, delivering a full-scale autonomous racecar operating reliably at up to 155 mph in multi-agent competition
- Led a 7-person subteam on state estimation, adding IMU-centric SLAM for GPS-denied robustness and EKF-based dead-reckoning failsafes for degraded sensing
- Built a unified ground-truth and track-artifact API used across perception, localization, planning, and control

**Robotics Institute Summer Scholars (RISS), Carnegie Mellon University (🔗)** — **06/2023–08/2023**
*Summer Scholar (Advisors: Sebastian Scherer, Micah Corah)* — *Pittsburgh, PA*

- Built a photorealistic IsaacSim simulator and a minimum-snap controller for aggressive quadrotor flight, supporting a learned collision-free navigation planner for multi-drone filming

**ESTAT Actuation (Startup R&D) (🔗)** [Media] — **01/2023–12/2023**
*Lead Software Developer & Mechanical Co-op—Supervisor: Kirby Witte* — *Pittsburgh, PA*

- Built longevity, linear, and high-torque test stands to characterize electrostatic robotic clutches
- Developed a low-level motor controller with torque feedback in embedded C, streaming measurements to a multithreaded Python planner
- Overhauled software source control, validation testing, and documentation for the test-stand codebase

## Publications

[C1] **TartanDrive 3: Scaling Full-Stack Data Across Modalities, Behaviors, and Extreme Environments**
*In Submission, 2026.* **M. Nye**[*], M. Sivaprakasam[*], S. Sharma, G. Hemkumar, S. Triest, Z. Zhao, Y. Liu, P. Maheshwari, A. Yanovich, D. Atha, W. Wang, S. Scherer.

[C2] **Structured Perception Representations for Off-road Dynamics Learning**
*CoRL, 2026.* **M. Nye**, S. Triest, M. Sivaprakasam, I. Adu, S. Scherer, W. Wang.

[C3] **Distilling Global Traversability Priors for Image-based Affordance Prediction in Off-road Environments**
*IROS, 2026.* M. Sivaprakasam, S. Triest, **M. Nye**, D. Atha, S. Khattak, D. Fan, W. Wang, S. Scherer.
Paper | Website | Video

[C4] **A Quantitative Analysis of Undergraduate Researchers' Intent to Apply to Graduate School**
*IEEE World Engineering Education Conference, 2026.* R. Burcin, **M. Nye**, I. Adu, V. Mruthyunjaya, J. M. Dolan.
Paper

[C5] **BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy**
*ICRA, 2025.* **M. Nye**, A. Raji, A. Saba, E. Erlich, R. Exley, A. Goyal, A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, S. Scherer.
Paper | Website

[C6] **TartanDrive 2.0: More Modalities and Better Infrastructure to Further Self-Supervised Learning Research in Off-Road Driving Tasks**
*ICRA, 2024.* M. Sivaprakasam, P. Maheshwari, M. G. Castro, S. Triest, **M. Nye**, S. Willits, A. Saba, W. Wang, S. Scherer.
Paper | Website | Video | Code

[J1] **Fast and Modular Autonomy Software for Autonomous Racing Vehicles**
*Journal of Field Robotics, Vol. 4, 2024 (now retitled as IEEE Transactions on Field Robotics).* A. Saba, A. Adetunji, A. Johnson, A. Kothari, M. Sivaprakasam, J. Spisak, P. Bharatia, A. Chauhan, B. Duff Jr., N. Gasparro, C. King, R. Larkin, B. Mao, **M. Nye**, A. Parashar, J. Attias, A. Balciumas, A. Brown, C. Chang, M. Gao, C. Heredia, A. Keats, J. Lavariega, W. Muckleroy III, A. Slavescu, N. Stathas, N. Suvarna, C. T. Zhang, S. Scherer, D. Ramanan.
Paper

[J2] **Simulation and Control for Learning Collision-free Navigation for Aerial Vehicles**
*Robotics Institute of Summer Scholars Working Papers Journal, Vol. 11, 2023.* **M. Nye**, M. Corah, S. Scherer.
Paper | Video | Poster

[W1] **TartanDrive 1.5: Improving Large Multimodal Robotics Dataset Collection and Distribution**
*ICRA Workshop on Pretraining for Robotics, 2023.* M. Sivaprakasam, S. Triest, M. G. Castro, **M. Nye**, M. Maulimov, C. Ho, P. Maheshwari, W. Wang, S. Scherer.
Paper

*Equal contribution.

## Fellowships, Grants, & Awards

- CMU Rales Fellow (Inaugural Cohort) (**$191k**) (🔗) 04/2024 – 05/2026
- Lead author, "VISTA: Versatile, Intelligent, Self-supervised Terrain Autonomy", DSO (**$1M**) 08/2025
- NSF Graduate Research Fellowship Honorable Mention (🔗) 04/2024
- 1st Place (of 9), Startup Challenge at Automate 2023, with ESTAT Actuation (**$10k**) 05/2023
- NSF REU Scholar 08/2023
- CMU RISS Scholar (**$8k**) 08/2023
- Pitt Success Merit Grant (**$8k**) 08/2020–08/2024
- Pennsylvania Ready to Succeed Scholarship (**$6.6k**) 02/2023

## Leadership, Service, & Outreach

### Professional Service

- Reviewer for CoRL, RA-L, T-RO, ICRA, IROS, and workshops 10/2024–Present
- Co-led Organizer for IROS 2026 Workshop on Bridging Perspectives in Navigation (🔗) 09/2026
- Organizer for RSS 2025 Workshop on Resilient Off-road Autonomous Robotics (🔗) 06/2025

- Organizer for undergraduate Intro to ROS Robotics Workshop (🔗) 03/2024
- CMU RISS REU Admissions Committee 01/2024

### Leadership & Teaching

- Principal Investigator for Study on REU Impact 08/2025
- Researcher for Robotics Institute Demographics and Spatial Equity (🔗) 05/2024–01/2025
- Robotics Teacher at Summer Camp with United Way Venango County (🔗) 07/2024–08/2024
- Robotics Teacher with the Northside SAFE for the Summer Program (🔗) 06/2024–08/2024
- RISS Scholar Buddy 06/2024–08/2024
- Outreach Initiative Team Member for CMU RISS Robolaunch (🔗) 08/2023–05/2024
- Student Leader at Community Engagement Center Robotics Week (🔗) 04/2024
- Certified Tutor for Calculus & Physics at University of Pittsburgh Study Labs 08/2021–05/2022

## Technical Skills

**Research Areas**: Robot Learning, Learned Dynamics, Off-road Autonomy, Field Robotics, Multimodal Datasets
**Programming**: Python, C++, Embedded C, MATLAB
**Tools**: PyTorch, ROS 2, DDS, Docker, GTSAM, OpenCV, IsaacSim
**Hardware**: Full-scale field deployment, system integration, rapid prototyping (CAD, 3D printing, machining)